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AI-Powered Multilingual Healthcare Assistant: Voice and Text-Enabled Medical Chatbot

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***Abstract*— In today's fast-paced and technologically advanced world, the healthcare sector faces numerous challenges in providing immediate, dependable, and personalized assistance to patients, particularly when it comes to scheduling appointments and handling grievances. The currently existing solution in the market do not offer multilingual support, which makes it challenging to work with a variety of linguistic populations. The proposed AI Model which offers multilingual support for providing the home remedies in the form of both text-based interaction by Claude API. The voice-based interactions between human and AI using google speech recognition which uses Deep Neural Network (DNN), it extracts the features such as Mel-frequency cepstral coefficients (MFCCs). It provides the features such as doctor booking system based on GPS Location according to disease, Grievance management system, It can able to generate the answer for the question based on information in Medical Report uploaded by the user Our proposed solution is designed to fix the problems with the current system, thereby improving the healthcare sector thus it can be used by people of different backgrounds. The proposed solution also has an accuracy rate of 92%, which is higher than other models and gives people reliable performance.**

Keywords*— *AI chatbot, multilingual support, home remedies, grievance management, Report generation

I. INTRODUCTION

Healthcare is one of the most critical sectors in society, yet it faces several challenges, especially in providing accessible, personalized, and multilingual assistance to diverse populations. One of the key challenges includes the need for timely and accurate advice, appointment scheduling, and grievance management. Many healthcare platforms are mostly in English and have minimal language assistance, which puts a lot of people at a disadvantage. Additionally, voice-based interaction is low, even though many users, especially the elderly and less educated people, find voice-based communication to be more intuitive. Thus, there is an increasing need for AI-powered systems that can offer interactive, multilingual healthcare support, improving patient satisfaction and process efficiency.

Existing healthcare systems are inefficient in several ways: (1) Language Barriers: Many existing healthcare platforms are restricted to English or a few regional languages, which makes it difficult for non-English speakers to access and understand medical information. This leads to a gap in healthcare delivery, particularly for uneducated and rural people. (2) Voice Interaction: There is a lack of interactive voice-based solutions, leaving elderly or less literate users struggling to use healthcare systems that rely heavily on text input. (3) Complex Processes: Booking appointments, accessing medical advice, and filing grievances often involve navigating through multiple steps on online platforms. These processes can be challenging and confusing, leading to delays and dissatisfaction among patients. The proposed solution tries to overcome these challenges by developing an AI-driven chatbot capable of offering medical advice, booking appointments, and managing grievances through a simple, interactive, and multilingual interface.

The primary objectives of proposed solutions are:

1) Provide Medical Advice: The AI-based solution will offer users reliable medical advice, such as home remedies for minor ailments and personalized diet plans.

2) Multilingual Support: The system will be designed to support many Indian languages, ensuring inclusivity for diverse linguistic groups.

3) Interactive Appointment and Grievance Management: The chatbot will facilitate appointment booking with healthcare providers and allow users to file medical grievances in an interactive, conversational manner. This will streamline the processes, making them more user-friendly and efficient, while providing real-time feedback and updates to users.

II. LITERATURE SURVEY

In Babylon Health [1] It provides real-time, AI-driven medical advice and symptom checking an also offers seamless video consultations with doctors. The gaps identified are it does not support Indian languages, restricting accessibility for non-English speakers in India. Lacks the ability to analyse and respond to user queries based on uploaded medical documents. In MFine (India-based) [2] It was designed for the Indian healthcare system,

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offering medical advice through AI. Facilitates direct consultations with doctors and lab test services. It does not fully support the wide range of Indian languages. unable to provide AI-driven analysis or responses based on uploaded medical documents.

Healthcare Chatbot by Ayain John et.al, had proposed solution that describes a healthcare chatbot that uses AI and NLP to help users identify symptoms and access medical information. It assists in improving healthcare efficiency by supporting communication between patients and providers [2]. Personal Healthcare Chatbot for Medical Suggestions Using AI and ML by Jegadeesan Ramalingam et al.,[3] had proposed solution which presents a chatbot that predicts diseases based on symptoms using NLP. It offers home remedies for minor issues and recommends doctor consultations for serious cases, achieving 82% accuracy. BERT-Based Medical Chatbot by Arun Babu et al., had proposed a BERT-based medical chatbot that improves healthcare communication by understanding complex medical language [4]

Healthcare Chatbot System by Mark Lawrence et al., had proposed solution which presents an AI-powered healthcare chatbot that uses machine learning and NLP to diagnose symptoms, offer personalized solutions, and recommend doctors but the accuracy was less [5]. AI Based Healthcare Chatbot by Riddhi Shetty et al., had proposed solution an AI-based healthcare chatbot that provides 24/7 medical support by diagnosing symptoms, recommending treatments, and booking appointments. Using NLP and machine learning, the chatbot aims to improve healthcare accessibility, reduce doctor workload, and save time and costs for patients, especially during medical crises, but it was designed with limited languages support [6]. Chatbot for Healthcare System Using Artificial Intelligence by Lekha Athota et al., had proposed solution on healthcare chatbot that uses artificial intelligence (AI) to interpret n-gram, TF-IDF, and cosine similarity queries to offer basic medical assistance. By providing fast answers from a knowledge base or expert input, it improves healthcare accessibility and lowers costs [7]. Healthcare Chatbot System using Artificial Intelligence Nikita Vijay Shinde et aal., had proposed a healthcare chatbot that uses AI and natural language processing (NLP) to provide medical recommendations, symptom-based advice, and primary care guidance prior to seeing a doctor. It analyses queries and provides prompt, economical help using methods like TF-IDF, n-grams, and cosine similarity in [8].

Healthcare Chatbot by Athulya N et al. had proposed solution presents a healthcare chatbot that based on user symptoms it predicts diseases using a Decision Tree algorithm. It offers text and voice interactions. Available 24/7, the chatbot enhances healthcare accessibility and reduces costs, with an accuracy rate of 78.24% [9]. Revitalizing Traditional Health Practices with Healing Hands: An AI powered Chatbot by Jasjot Singh et al., had proposed an AI-powered chatbot that uses user profiles and symptoms to provide individualized health advice and evidence-based home treatments for a range of health issues. The design, functionality, and efficacy of the chatbot are highlighted, along with its potential to improve healthcare by fusing traditional care with natural therapies [10].

Chatbot for Disease Prediction and Treatment Recommendation using Machine Learning Rohit Binu Mathew et al., had proposed a medical chatbot that uses natural language processing and machine learning (KNN) to forecast illnesses and suggest remedies based on user symptoms [11]. Chatbot based Disease Prediction and Treatment Recommendation using AI by Mr. Chandramaprasad Pathak, Dr. Namrata Ansari This proposed solution introduces a chatbot system for disease prediction and treatment recommendations using natural language processing and Artificial Intelligence. By analysing user- reported symptoms, it predicts potential diseases and suggests treatments [12].

Chatbots in Healthcare by Velibor Bozic - The proposed solution presents a chatbot system that employs natural language processing and Artificial Intelligence for disease prediction and treatment recommendations. By analyzing user-reported symptoms, it predicts. potential conditions and suggests treatments [13].

Machine Learning Based Healthcare Chatbot Rhythm Goel et al., had proposed a healthcare chatbot using natural language processing (NLP) and neural networks to diagnose illnesses, give users basic health information, and advise them when to see a doctor. Through a private and intuitive interface, this approach seeks to improve accessibility, lower healthcare costs, and solve user concerns [14].HealthCare Chatbot Using Machine Learning and NLP by Ronak Surve et al., had proposed a healthcare chatbot using natural language processing (NLP) and neural networks to detect illnesses, offer health advice, and suggest when to see a doctor. It seeks to lower expenses, increase accessibility to healthcare, and provide users with private support [15].

III. PROPOSED SYSTEM

The Smart Healthcare Assistant is an advanced AI-powered chatbot designed to address critical challenges in the healthcare domain by offering key functionalities such as medical advice, appointment booking, grievance management, and document-based question answering. It supports multilingual interaction and provides a user-friendly interface, ensuring inclusivity for users with diverse linguistic and technical abilities. The system is enhanced by cutting-edge technologies such as Natural Language Processing (NLP) and voice-based interaction to enhance accessibility, particularly for users with limited literacy or technical expertise. By addressing the shortcomings of existing healthcare systems, this solution provides a conversational, interactive platform that boosts efficiency, accessibility, and patient satisfaction. The figure 1 illustrate proposed system works and depicts the overall architectural flow of the system and provide detail structural overview of the proposed system. The mode of response for query given by the users are chat message or voice message.

A. Text and Voice Interaction

The systems enable users to interact with the chatbot using text or voice inputs.

In Text-Based Interaction, users can type their queries in their preferred and known comfortable language. The preprocessing of the input data consists of eliminating excessive spaces, special characters, or formatting errors is

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known as text cleaning. The process of tokenization involves dividing the text into smaller parts called tokens. Apply the AI Model for Text Processing (like Claude or GPT) receives the processed text and uses it to produce a response. The model generates text-based answers using methods such as neural networks or transformers. To produce coherent replies, a transformer weighs various elements of the incoming text utilizing self-attention processes while processing the complete text in simultaneously. The model can respond to queries, create dialogue, or provide specialized information since a significant amount of text data has been used to pre-train it and has learned the structure of language. The generated response typically goes through a decoding process, where the model predicts the most likely sequence of words or phrases that form a meaningful response.

In Voice-Based Interaction, the system processes voice inputs using speech recognition, converting them into text for analysis. Responses are delivered through TTS in the selected language, ensuring inclusivity for diverse user groups. Initially, speech is converted to text. This is done using a speech recognition model.

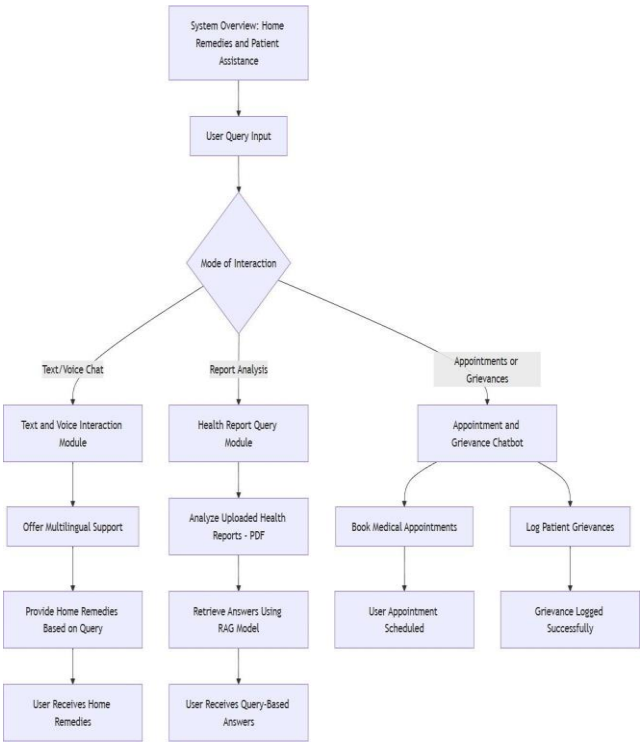

Figure 1. System Architecture of Multilingual AI Chat Bot

B. Report Analysis

In this module, User will upload pdf, then using Fitz function we will extract data from pdf, using graph-vect-rag library we will create a knowledge graph for the document and store in json format, also we will split the document in chunks and store it. Then while user interacting with the document, we will retrieve the context from both vector database and relationships context from knowledge graph and pass it to the LLM along with the user query, which will give us the final answer for the user question. Also, by using prompt engineering techniques for fine tuning the idea to generate content only from the user uploaded documents related to healthcare domain alone

C. Appointment Booking and Grievance Chatbot

This module simplifies appointment scheduling and grievance handling. In Appointment Booking system, the users can view a list of specialized doctors, check their availability, and book appointments through an interactive interface. In Grievance Management the chatbot allows users to submit grievances related to healthcare services, tracks their status, and provides updates in real time, ensuring a transparent process. The Follow of the system in which user gives the input in natural language. It is Pre-processed and vectorize the text input.

a) User's intent is classified using a model based on machine learning.

b) Generate a relevant response based on the classified intent.

c) Allow the chatbot model to be saved and can be reused through file management.

This module incorporates NLP techniques such as text preprocessing, tokenization, vectorization, and intent classification to understand user queries and to provide intelligent responses

D. Preprocessing and Feature Extraction

In Audio Preprocessing, the raw audio is processed to remove noise, normalize volume, and detect speech boundaries.

After the pre-processed the audio input, Feature Extraction is to be carried out to extract the essential features. It uses Google Speech Recognition API for Speech-to- Text conversion of the data. It Utilizes Deep Neural Networks (DNNs) and sequence-to-sequence models with attention mechanisms. It transforms speech waveforms into text by extracting features like Mel-frequency cepstral coefficients (MFCCs) to extract phonetic and spectral features of speech and also extract the The mel – spectrograms features with combination of time and frequency representation of the audio signals. Here X represents the audio frames in terms of time and Y represents the frequency in terms of mel scale. A trained speech-to-text model processes the features and converts the audio into text.

E. Algorithm

Step 1: User Input Handling

Detect input mode (Text or Voice)

If (mode ==Text): then accept user input as text

Else if (mode ==Voice):

then Capture speech and convert speech to text using Google Speech Recognition API Return user input

Step 2: Preprocess Text(text):

Step 2.1: Clean and preprocess the text by removing noise and special characters

Step 2.2: The raw text is split into tokens. Token is converted to Numerical representation by word embedding

Step 3:AI-Powered Text Response Generation.

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Step 3.1: Pass the processed text to the Claude API, as it uses transformer-based AI Model

Step 3.2: Model calculates relationship between the token using self-Attention

Step 3.3: Assign the scores for each token relative to others

Step3.4: Attention weighted embedding is passed to deep neural networks.

Step 3.5: Activation functions are applied such as ReLu are applied

Step 3.6: The models repeat the attention and feed forward mechanism to generate a meaningful, context-aware response.

Step 4: Multilingual Translation: Convert text input and output into the user's preferred language using the Google Translator API which in terms uses Neural Machine Translation (NMT) model

Step 5: Voice Response Handling (For Voice-Based Interactions), Convert the chatbot's text response into speech using gTTS (Google Text-to-Speech API). Step 6: Play the generated audio response for the user.

This algorithm shows the core functionalities of the medical chatbot, including multilingual support, and response for the query. It ensures seamless interaction through both text and voice input, enhancing accessibility and user experience for the users.

IV. RESULTS AND DISCUSSION

The implementation of the Smart Healthcare Assistant focused on the text-based and voice-based chat modules, achieving notable outcomes in enhancing accessibility and user interaction.

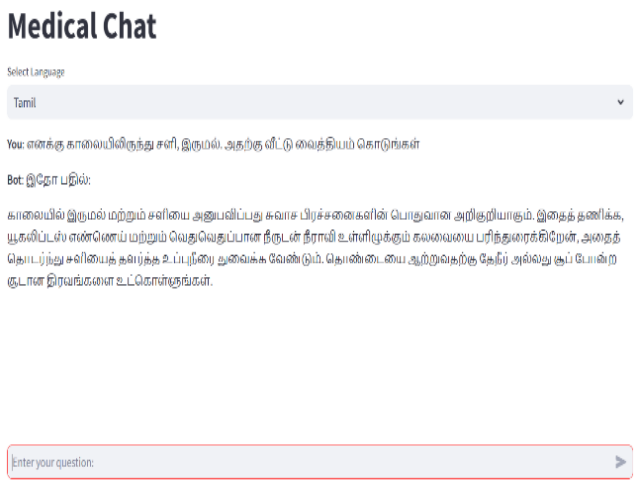

Figure 2. Text Chat Query Result

In Figure 2, the text-based chatbot provides relevant and accurate responses to medical queries, including home remedies. The translation feature ensured inclusivity for non-English speakers.

Figure 3: Voice Chat Query Result

In Figure 3, voice chat accurately converted the spoken inputs into text for processing and provides relevant and accurate responses to medical queries in the form of audio and text, enabling hands-free interaction. It can be used by non-English speakers.

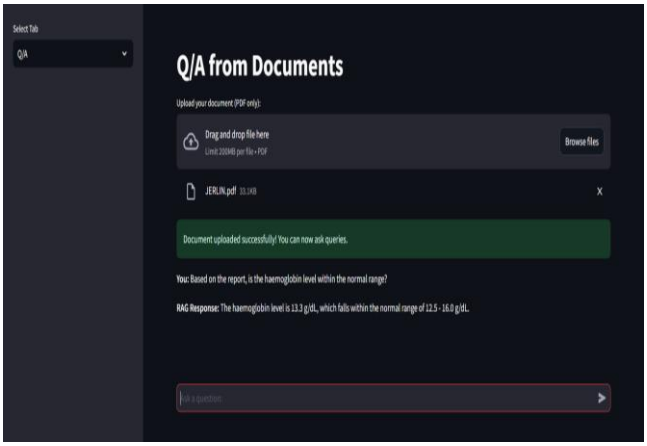

Figure 4: Report Analysis

In Figure 4, the system extracts text from PDFs and splits those extracted text into chunks and creates a knowledge graph. When user queries, it retrieves relevant context from both source from PDF and generates healthcare-specific responses using a Large Language Model that matches the user query.

Appointment booking and grievance booking leverages machine learning models for intent recognition and response generation; thus, it makes it a suitable solution for building an interactive conversational agent. Report Analysis Module allows users to upload a PDF document. The system will extract the relevant data from the PDF, break it into smaller parts, and for document's contents it creates a structured knowledge map. In addition to this, the system will be fine-tuned to focus only on content related to healthcare, ensuring that the answers are relevant to the user's specific needs in that domain.

B. Comparison of Model's Performance

In Figure 5, the chart compares the accuracy, Precision, Recall, F1-Score of four healthcare chatbot models, including three existing models and the proposed system developed in this research. The models compared are

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Babylon, Healthcare Chatbot, Personal Healthcare Chatbot for Medical Suggestions, and the Proposed System.

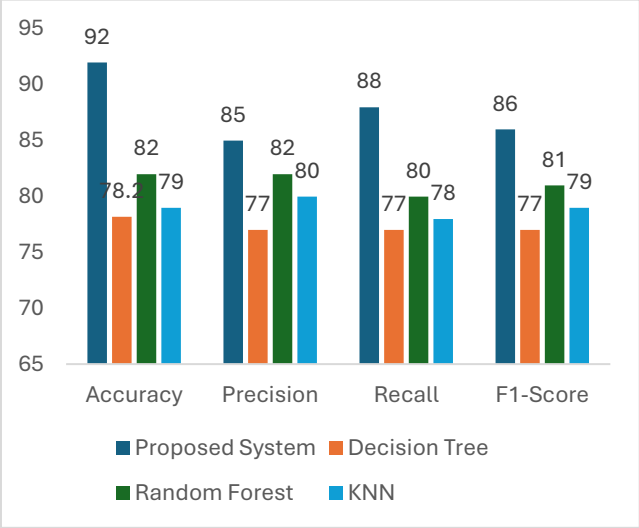

Figure.5. Performance Metrics comparison of Healthcare Chatbot Models

As demonstrated, the Proposed System outperforms the current models by achieving the greatest accuracy of 92%, 85% of Precision, 88% of Recall and 86% of F1-Score. The Healthcare Chatbot model using decision tree algorithm has a slightly lower accuracy rating of 78.24% , 75% of Precision, 77% of Recall and 76% of F1-Score than the other models, with random forest having an accuracy rate of 82%, 80% of Precision, 81% of Recall and 80% of F1-Score and Personal Healthcare Chatbot using K-Nearest Neighbour having an accuracy rate of 82%, 79% of Precision, 80% of Recall and 79% of F1-Score. According to the statistics, the suggested system provides a higher degree in each metrics and is a more dependable answer to medical questions and chatbot interactions in the healthcare industry.

V. CONCLUSION

The proposed solution Smart Healthcare Assistant developed addresses many challenges in the healthcare domain and services, including language barriers, lack of voice interaction, and inefficiencies in appointment booking and grievance management. By integrating Natural Language Processing (NLP), multilingual support, and voice-based interaction, the system provides more user-friendly solutions for diverse people. Through features such as home remedies, appointment scheduling, grievance management, and the feature to analyse medical reports, the Smart Healthcare Assistant provides an overall user-friendly healthcare experience for users, especially for those who face barriers due to language or literacy. The proposed system not only fills gaps left by existing healthcare solution but also enhances accessibility and efficiency. Its ensures that the patients, irrespective of their language or technical knowledge, can access healthcare support seamlessly.

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